

DS ACTIVITY 2

DSA0404-FDS

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ACT1:

```
ds act1.py - C:/Users/Khaviya KL/OneDrive/Desktop/DS activity2/ds act1.py (3.13.5)
File Edit Format Run Options Window Help
import pandas as pd
import numpy as np

# Sample data with missing entries
data = {
    'Name': ['Alice', 'Bob', 'Charlie', 'David'],
    'Age': [25, np.nan, 30, 22],
    'Salary': [50000, 60000, np.nan, 45000]
}

df = pd.DataFrame(data)

# 1. Check for missing values
print("Missing count:\n", df.isnull().sum())

# 2. Fill missing Age with the column median
df['Age'] = df['Age'].fillna(df['Age'].median())

# 3. Drop rows where Salary is missing
df = df.dropna(subset=['Salary'])

print("\nCleaned DataFrame:\n", df)
```

```
IDLE Shell 3.13.5
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/Khaviya KL/OneDrive/Desktop/DS activity2/ds act1.py =====
Missing count:
Name      0
Age        1
Salary     1
dtype: int64

Cleaned DataFrame:
   Name  Age  Salary
0  Alice  25.0  50000.0
1   Bob   25.0  60000.0
3  David  22.0  45000.0

>>>
```

ACT2:

```
ds act2.py - C:/Users/Khaviya KL/OneDrive/Desktop/DS activity2/ds act2.py (3.13.5)
File Edit Format Run Options Window Help
import pandas as pd

data = {
    'Employee': ['Amy', 'John', 'Raj', 'Kevin', 'Sora'],
    'Department': ['HR', 'IT', 'IT', 'Sales', 'IT'],
    'Experience': [3, 7, 1, 5, 6]
}

df = pd.DataFrame(data)

# Filter for employees in IT with more than 5 years of experience
filtered_df = df[(df['Department'] == 'IT') & (df['Experience'] > 5)]

print(filtered_df)
```

```
IDLE Shell 3.13.5
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/Khaviya KL/OneDrive/Desktop/DS activity2/ds act2.py =====
Missing count:
Name      0
Age        1
Salary     1
dtype: int64

Cleaned DataFrame:
   Name  Age  Salary
0  Alice  25.0  50000.0
1   Bob   25.0  60000.0
3  David  22.0  45000.0

>>>
===== RESTART: C:/Users/Khaviya KL/OneDrive/Desktop/DS activity2/ds act2.py =====
   Employee Department  Experience
1     John         IT             7
4     Sora         IT             6

>>>
```

ACT3:

```
ds act3.py - C:/Users/Khavya KL/OneDrive/Desktop/DS activity2/ds act3.py (3.13.5)
File Edit Format Run Options Window Help

import pandas as pd

data = {
    'Store': ['North', 'South', 'North', 'West', 'South', 'West'],
    'Sales': [1200, 1500, 900, 1100, 1700, 1300],
    'Employees': [5, 7, 5, 4, 8, 4]
}

df = pd.DataFrame(data)

# Group by Store and calculate total sales and average employees
summary = df.groupby('Store').agg({
    'Sales': 'sum',
    'Employees': 'mean'
}).reset_index()

print(summary)
```

```
IDLE Shell 3.13.5
File Edit Shell Debug Options Window Help

Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/Khavya KL/OneDrive/Desktop/DS activity2/ds act1.py =====
Missing count:
Name      0
Age       1
Salary    1
dtype: int64

Cleaned DataFrame:
   Name  Age  Salary
0  Alice  25.0  50000.0
1   Bob   25.0  60000.0
3  David  22.0  45000.0

>>>
===== RESTART: C:/Users/Khavya KL/OneDrive/Desktop/DS activity2/ds act2.py =====
Employee Department  Experience
1      John         IT           7
4      Sora         IT           6

>>>
===== RESTART: C:/Users/Khavya KL/OneDrive/Desktop/DS activity2/ds act3.py =====
Store Sales  Employees
0  North   2100         5.0
1  South   3200         7.5
2  West   2400         4.0

>>>
```

Ln: 27 Col: 0

ACT4:

```
ds act4.py - C:/Users/Khavya KL/OneDrive/Desktop/DS activity2/ds act4.py (3.13.5)
File Edit Format Run Options Window Help

import pandas as pd

# Create a sample dictionary with numeric and categorical data
data = {
    'Age': [25, 30, 35, 40, 45],
    'Salary': [50000, 60000, 75000, 90000, 110000],
    'Department': ['HR', 'IT', 'IT', 'Marketing', 'HR']
}

# Convert the dictionary into a Pandas DataFrame
df = pd.DataFrame(data)

# 1. Default describe() - Summary of numeric columns
print("--- Numeric Summary ---")
print(df.describe())

# 2. Categorical describe() - Summary of text/object columns
print("\n--- Categorical Summary ---")
print(df.describe(include='object'))
```

```
IDLE Shell 3.13.5
File Edit Shell Debug Options Window Help

>>>
===== RESTART: C:/Users/Khavya KL/OneDrive/Desktop/DS activity2/ds act1.py =====
Missing count:
Name      0
Age       1
Salary    1
dtype: int64

Cleaned DataFrame:
   Name  Age  Salary
0  Alice  25.0  50000.0
1   Bob   25.0  60000.0
3  David  22.0  45000.0

>>>
===== RESTART: C:/Users/Khavya KL/OneDrive/Desktop/DS activity2/ds act2.py =====
Employee Department  Experience
1      John         IT           7
4      Sora         IT           6

>>>
===== RESTART: C:/Users/Khavya KL/OneDrive/Desktop/DS activity2/ds act3.py =====
Store Sales  Employees
0  North   2100         5.0
1  South   3200         7.5
2  West   2400         4.0

>>>
===== RESTART: C:/Users/Khavya KL/OneDrive/Desktop/DS activity2/ds act4.py =====
--- Numeric Summary ---
           Age      Salary
count  5.000000   5.000000
mean   35.000000  77000.000000
std    7.905694   23874.672773
min    25.000000   50000.000000
25%    30.000000   60000.000000
50%    35.000000   75000.000000
75%    40.000000   90000.000000
max    45.000000  110000.000000

--- Categorical Summary ---
           Department
count              5
unique              3
top                HR
freq               2

>>>
```

Ln: 46 Col: 0

ACT5:

```
ds act5.py - C:/Users/Khavya KL/OneDrive/Desktop/DS activity2/ds act5.py (3.13.5)
File Edit Format Run Options Window Help

import pandas as pd

# Creating a sample dataset
data = {
    'Name': ['Alice', 'Bob', 'Charlie', 'David', 'Eva', 'Frank', 'Grace'],
    'Score': [85, 92, 78, 95, 88, 72, 91]
}

df = pd.DataFrame(data)

# 1. Get the top 5 rows (default behavior)
print(df.head())

# 2. Get the top 2 rows
print(df.head(2))

# 3. Get the bottom 3 rows
print(df.tail(3))
```

```
IDLE Shell 3.13.5
File Edit Shell Debug Options Window Help

>>>
===== RESTART: C:/Users/Khavya KL/OneDrive/Desktop/DS activity2/ds act2.py =====
Employee Department Experience
1 John IT 7
4 Sora IT 6
>>>
===== RESTART: C:/Users/Khavya KL/OneDrive/Desktop/DS activity2/ds act3.py =====
Store Sales Employees
0 North 2100 5.0
1 South 3200 7.5
2 West 2400 4.0
>>>
===== RESTART: C:/Users/Khavya KL/OneDrive/Desktop/DS activity2/ds act4.py =====
--- Numeric Summary ---
Age Salary
count 5.000000 5.000000
mean 35.000000 77000.000000
std 7.905694 23874.672773
min 25.000000 50000.000000
25% 30.000000 60000.000000
50% 35.000000 75000.000000
75% 40.000000 90000.000000
max 45.000000 110000.000000
--- Categorical Summary ---
Department
count 5
unique 3
top HR
freq 2
>>>
===== RESTART: C:/Users/Khavya KL/OneDrive/Desktop/DS activity2/ds act5.py =====
Name Score
0 Alice 85
1 Bob 92
2 Charlie 78
3 David 95
4 Eva 88
Name Score
0 Alice 85
1 Bob 92
Name Score
4 Eva 88
5 Frank 72
6 Grace 91
>>>
```

Ln: 61 Col: 0